



Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence & Data Science

AY: 2026-27

Class:	TE-AI&DS	Semester:	V
Course Code:	2015111	Course Name:	Statistics for Machine Learning and Data Science Lab

Name of Student:	Yash Vilas Karande
Roll No. :	17
Experiment No.:	8
Title of the Experiment:	Apply Clustering and Dimensionality Reduction Techniques for Pattern Discovery
Date of Performance:	09/09/20
Date of Submission:	23/09/20

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Performance	5	
Understanding	5	
Journal work and timely submission	10	
Total	20	

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Performance	4-5	2-3	1
Understanding	4-5	2-3	1
Journal work and timely submission	8-10	5-8	1-4

Checked by

Name of Faculty : **Dr. Tatwadarshi P.N.**

Signature :

Date :

Aim: To apply clustering and dimensionality reduction techniques to the Pima Indians Diabetes Dataset to discover hidden patterns and visualize relationships among patients.



Objectives

After completing this experiment, the student will be able to:

1. Apply clustering and dimensionality reduction techniques to discover patterns in a high-dimensional dataset.
2. Interpret the resulting clusters and reduced-dimensional representations using suitable evaluation measures and visualizations.

Tools Required

- Python 3.x
- Google Colab / Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

Dataset Details

Dataset: Pima Indians Diabetes Dataset

The dataset contains medical information for **768 female patients**. The Outcome variable indicates diabetes status.

The Outcome variable may be excluded during unsupervised learning and used later to compare the discovered clusters with the known diabetes categories.

Procedure

1. Load the Pima Indians Diabetes Dataset.
2. Select relevant numerical features for unsupervised analysis.



3. Handle invalid or missing values.
4. Standardize the selected features.
5. Apply K-Means clustering.
6. Select a suitable number of clusters using the Elbow Method.
7. Evaluate the clustering using the Silhouette Score.
8. Apply PCA to reduce the dimensionality of the dataset.
9. Visualize the observations in the reduced two-dimensional space.
10. Analyze the discovered clusters and interpret the results.

Description of the Experiment

Unsupervised learning identifies patterns in data without using a target variable during model development.

The experiment follows: High-Dimensional Dataset, Data Preprocessing, Feature Standardization, Clustering, Cluster Evaluation, Dimensionality Reduction, Visualization and Interpretation

Clustering groups similar observations, while dimensionality reduction represents high-dimensional data using fewer dimensions for visualization and analysis.

Detailed Description of Statistical Methods

K-Means Clustering

K-Means divides observations into k clusters by minimizing the distance between observations and their assigned cluster centroid.

The objective function is:

$$J = \sum_{j=1}^k \sum_{x_i \in C_j} \|x_i - \mu_j\|^2$$

where:

- C_j = cluster j



- μ_j = centroid of cluster j

Elbow Method

The Elbow Method evaluates different values of k using within-cluster sum of squares.

The value of k is selected by identifying the point where adding more clusters provides diminishing improvement.

Silhouette Score

The Silhouette Score evaluates how well an observation fits within its assigned cluster.

$$S = \frac{b-a}{\max(a,b)}$$

where:

- a = average distance within the same cluster
- b = average distance to the nearest other cluster

A higher score generally indicates better-defined clusters.

Principal Component Analysis

PCA transforms correlated variables into a smaller number of uncorrelated principal components.

The principal components are obtained from the covariance matrix using eigenvalues and eigenvectors.

The first principal components capture the maximum possible variance in the data.

PCA can be used to represent the dataset in two dimensions for visualization.

Implementation:



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[1]

✓ 2s

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score
from sklearn.decomposition import PCA
```

[2]

✓ 0s

```
url = "https://raw.githubusercontent.com/jbrownlee/Datasets/master/pima-indians-diabetes.data.csv"

columns = [
    "Pregnancies",
    "Glucose",
    "BloodPressure",
    "SkinThickness",
    "Insulin",
    "BMI",
    "DiabetesPedigreeFunction",
    "Age",
    "Outcome"
]

df = pd.read_csv(url, names=columns)

df.head()
```

✓

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age	Outcome
0	6	148	72	35	0	33.6	0.627	50	1
1	1	85	66	29	0	26.6	0.351	31	0
2	8	183	64	0	0	23.3	0.672	32	1
3	1	89	66	23	94	28.1	0.167	21	0
4	0	137	40	35	168	43.1	2.288	33	1

[3]

✓ 0s

```
print("Dataset Shape:", df.shape)

print("\nColumns:")
print(df.columns)

print("\nMissing Values:")
print(df.isnull().sum())
```

Dataset Shape: (768, 9)

Columns:
Index(['Pregnancies', 'Glucose', 'BloodPressure', 'SkinThickness', 'Insulin',
 'BMI', 'DiabetesPedigreeFunction', 'Age', 'Outcome'],
 dtype='object')

Missing Values:

Pregnancies	0
Glucose	0
BloodPressure	0
SkinThickness	0
Insulin	0
BMI	0
DiabetesPedigreeFunction	0
Age	0
Outcome	0

dtype: int64



[4]
✓ Os

```
features = [  
    "Pregnancies",  
    "Glucose",  
    "BloodPressure",  
    "SkinThickness",  
    "Insulin",  
    "BMI",  
    "DiabetesPedigreeFunction",  
    "Age"  
]  
  
X = df[features].copy()  
  
X.head()
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age
0	6	148	72	35	0	33.6	0.627	50
1	1	85	66	29	0	26.6	0.351	31
2	8	183	64	0	0	23.3	0.672	32
3	1	89	66	23	94	28.1	0.167	21
4	0	137	40	35	168	43.1	2.288	33

[5]
✓ Os

```
invalid_columns = [  
    "Glucose",  
    "BloodPressure",  
    "SkinThickness",  
    "Insulin",  
    "BMI"  
]  
  
X[invalid_columns] = X[invalid_columns].replace(0, np.nan)  
  
print("Missing values after replacing invalid zeros:")  
print(X.isnull().sum())
```

```
... Missing values after replacing invalid zeros:  
Pregnancies      0  
Glucose          5  
BloodPressure    35  
SkinThickness    227  
Insulin          374  
BMI              11  
DiabetesPedigreeFunction  0  
Age              0  
dtype: int64
```

[6]
✓ Os

```
for column in invalid_columns:  
    X[column] = X[column].fillna(X[column].median())  
  
print("Missing values after preprocessing:")  
print(X.isnull().sum())
```

```
... Missing values after preprocessing:  
Pregnancies      0  
Glucose          0  
BloodPressure    0  
SkinThickness    0  
Insulin          0  
BMI              0  
DiabetesPedigreeFunction  0  
Age              0  
dtype: int64
```



[7]
✓ Os

```
scaler = StandardScaler()  
  
X_scaled = scaler.fit_transform(X)  
  
print("Standardized data shape:", X_scaled.shape)
```

✓

Standardized data shape: (768, 8)

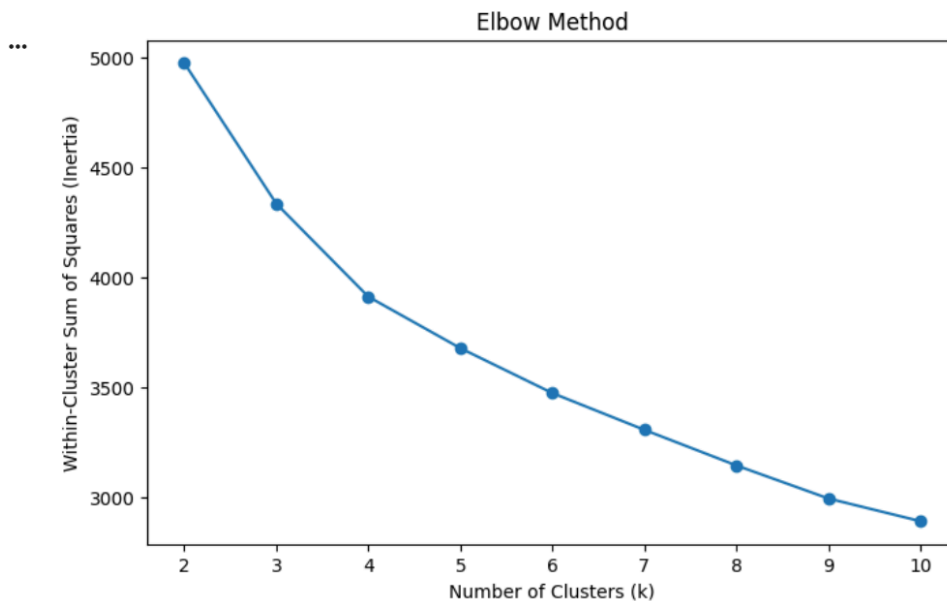
[8]
✓ Os

```
inertia = []  
  
for k in range(2, 11):  
    kmeans = KMeans(n_clusters=k, random_state=42, n_init=10)  
    kmeans.fit(X_scaled)  
    inertia.append(kmeans.inertia_)
```

[9]
✓ Os

```
plt.figure(figsize=(8, 5))  
  
plt.plot(range(2, 11), inertia, marker='o')  
  
plt.xlabel("Number of Clusters (k)")  
plt.ylabel("Within-Cluster Sum of Squares (Inertia)")  
plt.title("Elbow Method")  
  
plt.show()
```

✓





[10]

✓ 2s

```
silhouette_scores = []

for k in range(2, 11):
    kmeans = KMeans(n_clusters=k, random_state=42, n_init=10)
    labels = kmeans.fit_predict(X_scaled)

    score = silhouette_score(X_scaled, labels)
    silhouette_scores.append(score)

    print("k =", k, "Silhouette Score =", score)
```

✓

```
k = 2 Silhouette Score = 0.19627861483317974
k = 3 Silhouette Score = 0.19500407738247363
k = 4 Silhouette Score = 0.18979077032947686
k = 5 Silhouette Score = 0.17745305895865027
k = 6 Silhouette Score = 0.16816982746206624
k = 7 Silhouette Score = 0.12256077823302804
k = 8 Silhouette Score = 0.14631185790926513
k = 9 Silhouette Score = 0.14662635173937452
k = 10 Silhouette Score = 0.1448140925656229
```

[11]

✓ 0s

```
kmeans = KMeans(n_clusters=k, random_state=42, n_init=10)

clusters = kmeans.fit_predict(X_scaled)

print("Cluster labels:")
print(clusters[:20])
```

✓

```
... Cluster labels:
[2 0 8 0 6 8 1 9 5 7 8 2 2 5 7 9 4 9 1 1]
```

[12]

✓ 0s

```
df["Cluster"] = clusters

df.head()
```

✓

```
...
   Pregnancies  Glucose  BloodPressure  SkinThickness  Insulin   BMI  DiabetesPedigreeFunction  Age  Outcome  Cluster
0            6     148             72           35         0   33.6              0.627    50         1         2
1            1      85             66           29         0   26.6              0.351    31         0         0
2            8     183             64            0         0   23.3              0.672    32         1         8
3            1      89             66           23        94   28.1              0.167    21         0         0
4            0     137             40           35       168   43.1              2.288    33         1         6
```

[13]

✓ 0s

```
final_silhouette = silhouette_score(X_scaled, clusters)

print("Final Silhouette Score:", final_silhouette)
```

✓

```
... Final Silhouette Score: 0.1448140925656229
```

[14]

✓ 0s

```
pca = PCA(n_components=2)

X_pca = pca.fit_transform(X_scaled)

print("PCA shape:", X_pca.shape)
```

✓

```
PCA shape: (768, 2)
```



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[15]

✓ Os

```
print("Explained variance ratio:")
print(pca.explained_variance_ratio_)

print("\nTotal variance explained:")
print(pca.explained_variance_ratio_.sum())
```

▼

Explained variance ratio:
[0.28537316 0.18694722]

Total variance explained:
0.4723203741075307

[16]

✓ Os

```
loadings = pd.DataFrame(
    pca.components_.T,
    columns=["PC1", "PC2"],
    index=features
)

print(loadings)
```

▼

	PC1	PC2
Pregnancies	0.301434	0.557960
Glucose	0.423980	-0.082050
BloodPressure	0.377177	0.171980
SkinThickness	0.397103	-0.308746
Insulin	0.307134	-0.235780
BMI	0.401844	-0.398489
DiabetesPedigreeFunction	0.156874	-0.273482
Age	0.386013	0.517560

[17]

✓ Os

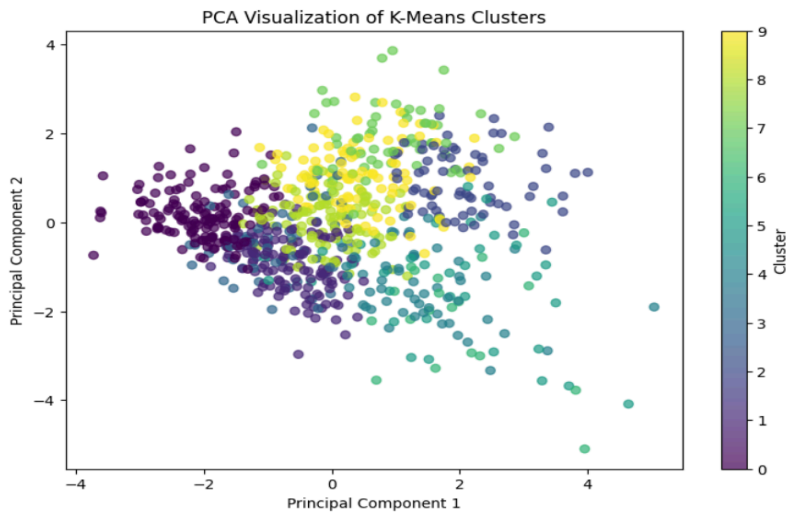
```
plt.figure(figsize=(9, 6))

plt.scatter(
    X_pca[:, 0],
    X_pca[:, 1],
    c=clusters,
    cmap="viridis",
    alpha=0.7
)

plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.title("PCA Visualization of K-Means Clusters")

plt.colorbar(label="Cluster")

plt.show()
```



[18]
✓ 0s

```
comparison = pd.crosstab(  
    df["Cluster"],  
    df["Outcome"],  
    margins=True  
)  
  
print(comparison)
```

Outcome	0	1	All
Cluster			
0	141	4	145
1	95	29	124
2	16	54	70
3	40	16	56
4	28	29	57
5	15	20	35
6	6	13	19
7	36	22	58
8	61	47	108
9	62	34	96
All	500	268	768

[19]
✓ 0s

```
cluster_outcome = pd.crosstab(  
    df["Cluster"],  
    df["Outcome"],  
    normalize="index"  
) * 100  
  
print(cluster_outcome)
```

Outcome	0	1
Cluster		
0	97.241379	2.758621
1	76.612903	23.387097
2	22.857143	77.142857
3	71.428571	28.571429
4	49.122807	50.877193
5	42.857143	57.142857
6	31.578947	68.421053
7	62.068966	37.931034
8	56.481481	43.518519
9	64.583333	35.416667



Conclusion:

Clustering and dimensionality reduction techniques were applied to the Pima Indians Diabetes Dataset. K-Means clustering was used to group patients based on similarities in their medical attributes, while PCA was used to reduce the dimensionality of the dataset for visualization. The resulting clusters were evaluated and interpreted using appropriate measures and visualizations.

Questions to be Answered by Students

1. What value of k was selected using the Elbow Method, and why?

ANS: The value of $k = 3$ was selected using the Elbow Method. At $k = 3$, the reduction in within-cluster sum of squares (inertia) begins to decrease more gradually with further increases in the number of clusters. Therefore, $k = 3$ provides a suitable balance between the number of clusters and within-cluster variation.

2. What Silhouette Score was obtained, and what does it indicate about the quality of the clusters?

ANS: The Silhouette Score obtained for the selected clustering was 0.1448140925656229. The score indicates how well the observations are grouped within their assigned clusters and how well separated the clusters are from each other. A higher positive value indicates better-defined and more separated clusters.

3. Which variables contributed most to the first two principal components?

ANS: The contribution of each variable was determined using the absolute PCA loading values.

- **PC1:** The variables contributing most are **Glucose (0.4240)**, **BMI (0.4018)**, **SkinThickness (0.3971)**, **Age (0.3860)**, and **BloodPressure (0.3772)**.
- **PC2:** The variables contributing most are **Pregnancies (0.5580)**, **Age (0.5176)**, and **BMI (0.3985)**, followed by **SkinThickness (0.3087)**.

Therefore, **Glucose has the highest contribution to PC1**, while **Pregnancies has the highest contribution to PC2**.



4. What patterns were observed in the PCA visualization of the clusters?

ANS: The PCA visualization showed that the observations formed groups corresponding to the K-Means clusters, although there was some overlap between the clusters. This indicates that the patient data contains some differences in their medical characteristics, but the clusters are not completely separated in the two-dimensional PCA space.

5. How do the discovered clusters compare with the known Outcome categories?

ANS: The K-Means clusters were formed without using the Outcome variable. When the clusters were compared with the known Outcome categories, some clusters contained a higher proportion of diabetic patients while others contained a higher proportion of non-diabetic patients. However, the clusters did not perfectly match the Outcome categories, indicating that the medical attributes used for clustering do not completely separate diabetic and non-diabetic patients.